

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Large pre-trained language models have been shown to store factual knowledge in their parameters. However, their ability to access and precisely manipulate knowledge is still limited. We build Retrieval-Augmented Generation (RAG) models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever.

Introduction

Pre-trained neural language models learn a substantial amount of in-depth knowledge from data. However, they have drawbacks: they cannot easily expand or revise their memory, they can struggle to provide insights into their decisions, and they frequently produce hallucinations when generating factual statements.

Methodology

We introduce RAG models, which combine parametric and non-parametric memory. Our non-parametric component is a Dense Passage Retriever (DPR) querying a vector index of Wikipedia with 21M documents. The parametric component is BART-large (400M parameters). We evaluate two models: RAG-Sequence (which uses the same retrieved document to generate the entire sequence) and RAG-Token (which can use different retrieved documents for each token).

Experiments & Evaluation

We test on four open-domain Question Answering datasets: Natural Questions (NQ), TriviaQA, WebQuestions (WQ), and CuratedTREC, as well as Jeopardy question generation and MS-MARCO.

Results & Discussion

RAG models set new state-of-the-art results on open-domain QA benchmarks, outperforming parametric-only models like T5 11B while using significantly fewer parameters. In human evaluations on factuality and specificity, RAG generations are rated as more factual than pure parametric baselines.

Limitations

RAG performance is fundamentally bounded by retriever recall. If relevant context is not retrieved in the top-K candidates, the generator may default to hallucinations or parametric uncertainty. Furthermore, dense retrieval across millions of passages incurs indexing and latency overheads.

Conclusion & Future Work

We have presented RAG models that fuse parametric generative capabilities with non-parametric vector retrieval. We demonstrate strong empirical benefits across diverse knowledge-intensive benchmarks.